

1) Настройка локальных сетей с использованием протокола IEEE 802.1Q

Первым делом создадим сети VLAN 333 и 20.

В терминале коммутатора Layer2Switch-1:

```
conf t
  vlan 333
    name VLAN333

  vlan 20
    name VLAN20
end
write
```

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 333
Switch(config-vlan)#name VLAN333
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#end
Switch#write
```

Для проверки созданных сетей.

```
show vlan brief
```

```
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi2/0
20   VLAN20                  active
100  VLAN100                 active
200  VLAN0200                active
300  VLAN0300                active
333  VLAN333                  active
1002 fddi-default            act/unsup
1003 trcrf-default         act/unsup
1004 fddinet-default        act/unsup
1005 trbrf-default         act/unsup
Switch#
```

Для начала посмотрим к каким конкретно портам коммутатора Layer2Switch-1 подключены какие коммутаторы.

```
show mac address-table
```

```
Switch>show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0c05.c04d.0000    DYNAMIC Gi1/0
1       0c50.16ac.0000    DYNAMIC Gi0/0
1       0c50.16ac.0001    DYNAMIC Gi0/1
1       0c83.f4f1.0000    DYNAMIC Gi1/2
1       0c83.f4f1.0001    DYNAMIC Gi1/3
1       0ccd.f948.0000    DYNAMIC Gi0/2
1       0ccd.f948.0001    DYNAMIC Gi0/3
Total Mac Addresses for this criterion: 7
```

Также, определяем мак-адреса всех коммутаторов:

В терминале каждого коммутатора:

```
show interface
```

Во второй строке вывода находим мак-адрес:

Layer2Switch-1:

```
Switch#show interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Hardware is iGbE, address is 0c3d.a366.0000 (bia 0c3d.a366.0000)
```

Layer2Switch-2:

```
Switch#show interface
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is iGbE, address is 0c50.16ac.0000 (bia 0c50.16ac.0000)
```

Layer2Switch-3:

```
Switch>show interface
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is iGbE, address is 0ccd.f948.0000 (bia 0ccd.f948.0000)
```

Layer2Switch-4:

```
Switch>show interface
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is iGbE, address is 0c05.c04d.0000 (bia 0c05.c04d.0000)
```

Layer2Switch-5:

```
Switch>show interface
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is iGbE, address is 0c83.f4f1.0000 (bia 0c83.f4f1.0000)
```

Раз необходимо строить сеть по протоколу IEEE 802.1Q, значит используем для vlan 333 и 20 одинаковые порты. Но vlan 333 должен использовать Native VLAN.

```
conf t
  interface Gi0/2 # Порт с линком до Layer2Switch-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk allowed vlan 20,333
    switchport trunk native vlan 333
    no shutdown
    exit
end

  interface Gi1/0 # Порт с линком до Layer2Switch-4
    switchport trunk encapsulation dot1q
```

```

        switchport mode trunk
        switchport trunk allowed vlan 20,333
        switchport trunk native vlan 333
        no shutdown
        exit
    end

    interface Gi1/2 # Порт с линком до Layer2Switch-5
        switchport trunk encapsulation dot1q
        switchport mode trunk
        switchport trunk allowed vlan 20,333
        switchport trunk native vlan 333
        no shutdown
        exit
    end
write

```

Проверяем, что всё настроено верно

```
show interface trunk
```

Видим, что порты Gi0/2, Gi1/0 и Gi1/2 передают кадры VLAN 333 без тега, и, при этом передают кадры VLAN 20 с тегом:

```
Switch#show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	desirable	n-isl	trunking	1
Gi0/1	desirable	n-isl	trunking	1
Gi0/2	on	802.1q	trunking	333
Gi0/3	desirable	n-isl	trunking	1
Gi1/0	on	802.1q	trunking	333
Gi1/1	desirable	n-isl	trunking	1
Gi1/2	on	802.1q	trunking	333
Gi1/3	desirable	n-isl	trunking	1

Port	Vlans allowed on trunk
Gi0/0	1-4094
Gi0/1	1-4094
Gi0/2	20,333
Gi0/3	1-4094
Gi1/0	20,333
Gi1/1	1-4094
Gi1/2	20,333
Gi1/3	1-4094

Оставшиеся порты коммутатора переводим в режим access, и присваиваем vlan 1, чтобы избежать петель без протокола STP:

```

conf t
    interface GiX/Y

```

```
switchport mode access
switchport access vlan 1
shutdown
exit
end
```

Проверяем:

```
show interface trunk
```

```
Switch-1#show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/2	on	802.1q	trunking	333
Gil/0	on	802.1q	trunking	333
Gil/2	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/2	20,333
Gil/0	20,333
Gil/2	20,333

Теперь все порты Layer2Switch-2 переведём в режим access, и присвоим vlan 1:

Вывод всех портов коммутатора:

```
show interface status
```

```
Switch-2#show interface status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	1	auto	auto	unknown
Gi0/1		connected	1	auto	auto	unknown
Gi0/2		connected	1	auto	auto	unknown
Gi0/3		connected	1	auto	auto	unknown
Gil/0		connected	trunk	auto	auto	unknown
Gil/1		connected	trunk	auto	auto	unknown
Gil/2		connected	trunk	auto	auto	unknown
Gil/3		connected	trunk	auto	auto	unknown
Gi2/0		notconnect	1	auto	auto	unknown

Вывод всех портов коммутатора в режиме trunk:

```
show interface trunk
```

Вот эти порты и будем править:

```
Switch-2#show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gil/0	desirable	n-isl	trunking	1
Gil/1	desirable	n-isl	trunking	1
Gil/2	desirable	n-isl	trunking	1
Gil/3	desirable	n-isl	trunking	1

Port	Vlans allowed on trunk
Gil/0	1-4094
Gil/1	1-4094
Gil/2	1-4094
Gil/3	1-4094

```
conf t
    interface G1X/Y
        switchport mode access
        switchport access vlan 1
        shutdown
        exit
    end
write
```

Проверим:

```
show interface trunk
```

У коммутатора Layer2Switch-2 нет портов в режиме trunk:

```
Switch-2#show interface trunk
Switch-2#
```

Настроим коммутаторы Layer2Switch-3-5

Теперь настроим коммутатор Layer2Switch-3.

Для начала дадим всем коммутаторам осмысленные названия, чтобы было проще ориентироваться:

```
conf t
    hostname Switch-*номер коммутатора*
```

Выведем список соседних с Layer2Switch-2 коммутаторов, и порты, с которыми соединён коммутатор 3 с помощью протокола cdp:

```
show cdp neighbors
```

Коммутатор Layer2Switch-3 соединён с Layer2Switch-1 через порт Gi0/0:

```
Switch-3>show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID      Local Intrfce   Holdtme    Capability  Platform  Port ID
Switch-1       Gig 0/0        178        R S I      IOSv      Gig 0/2

Total cdp entries displayed : 1
```

Настроим порт Gi0/0:

```
conf t
  interface Gi0/0
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk allwed vlan 20,333
    switchport trunk native vlan 333
    no shutdown
    exit
end
write
```

Выведем статус всех портов Layer2Switch-3:

```
show interface status
```

Видим подключение в режиме trunk к коммутатору 1. Также видим три отключенных порта в vlan 1: два из них - подключение к коммутатору 2, один - к комутатору 1:

```
Switch-3>show interface status

Port      Name      Status      Vlan      Duplex  Speed  Type
Gi0/0     Name      connected   trunk     auto    auto   unknown
Gi0/1     Name      disabled    1         auto    auto   unknown
Gi0/2     Name      disabled    1         auto    auto   unknown
Gi0/3     Name      disabled    1         auto    auto   unknown
Gil/0     Name      connected   unassigned auto    auto   unknown
Gil/1     Name      connected   1         auto    auto   unknown
Switch-3>
```

Соответственно, оставшиеся два подключены к PC1 и PC2.

Теперь нужно понять кто из них кто. Для этого настроим оставшиеся два коммутатора, и будем присваивать компьютеры разным vlan, и пытаться пинговать компьютеры из этой же сети.

Настраиваем Layer2Switch-4 и Layer2Switch-5 по аналогии с Layer2Switch-3.

P.S. Я проделал действия в неправильном порядке, и не посмотрел к каким портам подключены компьютеры. Поэтому, пришлось поступить иначе, и подключить компьютеры

PC1-4 к локальным сетям наугад, а потом пинговать сначала с PC1 PC3 и PC4, а потом с PC2, и смотреть через show mac address-table в правильных ли сетях находятся компьютеры.

По итогу:

```
show interface status
```

Layer2Switch-3:

```
Switch-3>show interface status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		disabled	1	auto	auto	unknown
Gi0/2		disabled	1	auto	auto	unknown
Gi0/3		disabled	1	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

Layer2Switch-4:

```
Switch-4#show interface status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		disabled	1	auto	auto	unknown
Gi0/2		disabled	1	auto	auto	unknown
Gi0/3		disabled	1	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

Layer2Switch-5:

```
Switch-5#show interface status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		disabled	1	auto	auto	unknown
Gi0/2		disabled	1	auto	auto	unknown
Gi0/3		disabled	1	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

IP-адреса компьютеров:

PC1:

```
PC1> show ip
```

NAME	: PC1[1]
IP/MASK	: 192.168.0.1/24
GATEWAY	: 0.0.0.0
DNS	:
MAC	: 00:50:79:66:68:00

PC2:

```
PC2> show ip
```

```
NAME       : PC2[1]  
IP/MASK    : 192.168.0.2/24  
GATEWAY    : 0.0.0.0  
DNS        :  
MAC        : 00:50:79:66:68:01
```

PC3:

```
PC3> show ip
```

```
NAME       : PC3[1]  
IP/MASK    : 192.168.0.3/24  
GATEWAY    : 0.0.0.0  
DNS        :  
MAC        : 00:50:79:66:68:02
```

PC4:

```
PC4> show ip
```

```
NAME       : PC4[1]  
IP/MASK    : 192.168.0.4/24  
GATEWAY    : 0.0.0.0  
DNS        :  
MAC        : 00:50:79:66:68:03
```

PC5:

```
PC5> show ip
```

```
NAME       : PC5[1]  
IP/MASK    : 192.168.0.5/24  
GATEWAY    : 0.0.0.0  
DNS        :  
MAC        : 00:50:79:66:68:04
```

PC6:

```
PC6> show ip
```

```
NAME       : PC6[1]  
IP/MASK    : 192.168.0.6/24  
GATEWAY    : 0.0.0.0  
DNS        :  
MAC        : 00:50:79:66:68:05
```

2) Пинг всех PC

PC1:

```
PC1> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=23.054 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=4.801 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=2.975 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=9.918 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=2.668 ms

PC1> ping 192.168.0.4

host (192.168.0.4) not reachable

PC1> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=9.099 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=4.069 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=1.762 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=5.038 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=13.490 ms

PC1> ping 192.168.0.6

host (192.168.0.6) not reachable
```

PC2:

```
PC2> ping 192.168.0.3

host (192.168.0.3) not reachable

PC2> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=8.191 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=11.910 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=14.107 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=7.577 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=10.856 ms

PC2> ping 192.168.0.5

host (192.168.0.5) not reachable

PC2> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=9.748 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=8.731 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=7.525 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=16.445 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=12.919 ms
```

PC3:

```
PC3> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=7.069 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=10.968 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=11.008 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=8.085 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=15.995 ms

PC3> ping 192.168.0.2

host (192.168.0.2) not reachable

PC3> ping 192.168.0.3

192.168.0.3 icmp_seq=1 ttl=64 time=0.001 ms
192.168.0.3 icmp_seq=2 ttl=64 time=0.001 ms
192.168.0.3 icmp_seq=3 ttl=64 time=0.001 ms
192.168.0.3 icmp_seq=4 ttl=64 time=0.001 ms
192.168.0.3 icmp_seq=5 ttl=64 time=0.001 ms

PC3> ping 192.168.0.4

host (192.168.0.4) not reachable

PC3> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=6.072 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=5.057 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=16.248 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=6.312 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=5.697 ms

PC3> ping 192.168.0.6

host (192.168.0.6) not reachable
```

PC4:

```
PC4> ping 192.168.0.1

host (192.168.0.1) not reachable

PC4> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=14.907 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=2.634 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=10.329 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=12.475 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=7.698 ms

PC4> ping 192.168.0.3

host (192.168.0.3) not reachable

PC4> ping 192.168.0.5

host (192.168.0.5) not reachable

PC4> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=11.620 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=5.758 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=6.716 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=13.375 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=9.059 ms
```

PC5:

```
PC5> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=5.273 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=16.089 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=10.200 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=7.029 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=22.462 ms

PC5> ping 192.168.0.2

host (192.168.0.2) not reachable

PC5> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=3.656 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=31.439 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=10.488 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=7.021 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=19.167 ms

PC5> ping 192.168.0.4

host (192.168.0.4) not reachable

PC5> ping 192.168.0.5

192.168.0.5 icmp_seq=1 ttl=64 time=0.001 ms
192.168.0.5 icmp_seq=2 ttl=64 time=0.001 ms
192.168.0.5 icmp_seq=3 ttl=64 time=0.001 ms
192.168.0.5 icmp_seq=4 ttl=64 time=0.001 ms
192.168.0.5 icmp_seq=5 ttl=64 time=0.001 ms

PC5> ping 192.168.0.6

host (192.168.0.6) not reachable
```

PC6:

```
PC6> ping 192.168.0.1

host (192.168.0.1) not reachable

PC6> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=18.835 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=5.602 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=5.071 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=13.006 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=7.322 ms

PC6> ping 192.168.0.3

host (192.168.0.3) not reachable

PC6> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=7.791 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=6.217 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=10.939 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=12.998 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=9.546 ms

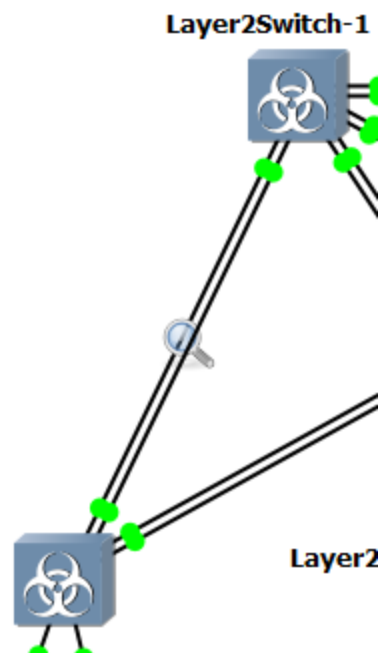
PC6> ping 192.168.0.5

host (192.168.0.5) not reachable
```

Компьютеры из сети 20 не могут пингануть компьютеры из сети 333, и наоборот.
Компьютеры из своей сети все компьютеры пингануть могут.

3) Перехват тегированных и нетегированных пакетов

С этого линка перехватываем пакеты:



Отфильтровываем пакеты vlan 20:

```
vlan.id == 20
```

vlan.id == 20						
No.	Time	Source	Destination	Protocol	Leng	Info
873	671.796841	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
875	673.806714	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
879	675.811013	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
881	677.823931	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
884	679.833709	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
886	681.841222	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
888	683.885064	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
891	685.913614	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
893	687.932358	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
896	689.945258	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
898	691.945797	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
902	693.959455	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
905	695.973188	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
907	697.976816	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
910	699.983573	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
912	701.992370	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
915	704.013477	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
918	706.023609	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
920	708.033543	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
924	710.051713	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
926	712.061693	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
928	714.080100	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
931	716.123844	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
933	718.130093	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
936	720.138587	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
938	722.152557	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
941	724.189956	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/2
- Frame 361: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0 - Ethernet II, Src: 0c:3d:a3:66:00:02 (0c:3d:a3:66:00:02), Dst: PVST+ (01:00:0c:cc:cc:cd) 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20 - Logical-Link Control - Spanning Tree Protocol						

В самом низу есть строка "802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20", она указывает на то, что кадр тегированный.

Если отфильтровать пакеты vlan 333, то возникает вот это:

vlan.id == 333							
No.	Time	Source	Destination	Protocol	Leng	Info	
1922	1478.082110	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1925	1480.092098	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1926	1480.268703	0c:cd:f9:48:00:00	CDP/VTP/DTP/PagP/UDLD	DTP	62	Dynamic Trunk Protocol	
1928	1482.105673	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1930	1484.121388	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1933	1486.140021	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1935	1487.977755	0c:3d:a3:66:00:02	CDP/VTP/DTP/PagP/UDLD	DTP	62	Dynamic Trunk Protocol	
1936	1488.148787	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1939	1490.166593	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1941	1492.188249	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1944	1494.251950	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1947	1496.259721	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1949	1498.257321	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1952	1500.268820	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1954	1502.271698	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1956	1504.298364	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1959	1506.308019	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1961	1508.316507	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1964	1510.332343	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1965	1510.557303	0c:cd:cf:48:00:00	CDP/VTP/DTP/PagP/UDLD	DTP	62	Dynamic Trunk Protocol	
1967	1512.334249	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1969	1514.357990	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1972	1516.374890	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1974	1518.393011	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1975	1518.529175	0c:3d:a3:66:00:02	CDP/VTP/DTP/PagP/UDLD	DTP	62	Dynamic Trunk Protocol	
1978	1520.404280	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4
1980	1522.418428	0c:3d:a3:66:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:05:c0:4d:00:00	Cost = 4

Frame 1416: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0

Ethernet II, Src: 0c:3d:a3:66:00:02 (0c:3d:a3:66:00:02), Dst: PVST+ (01:00:0c:cc:cc:cd)

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 333

- 000. = Priority: Best Effort (default) (0)
- ...0 = DEI: Ineligible
- ... 0001 0100 1101 = ID: 333

Length: 50

Logical-Link Control

Spanning Tree Protocol

Пакеты vlan 333тоже являются тегированными. Я всё перепроверил (Вероятно, я чего-то недопонял), почитал в интернете, и наткнулся на информацию, что это специфика работы GNS3 - тэгировать все кадры, даже те, что указана, как native.

4) Конфигурации коммутаторов:

Layer2Switc-1.log

```
===== PuTTY log 2025.12.15 02:16:08 =====
sh run
Building configuration...

Current configuration : 5667 bytes
!
! Last configuration change at 18:42:47 UTC Sat Dec 13 2025
!
version 15.0
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
```

```
hostname Switch-1
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
!
vtp domain CISCO-vIOS
vtp mode transparent
!
!
!
ip cef
no ipv6 cef
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
vlan 10
name VLAN10
!
vlan 20
name VLAN20
!
vlan 100
name VLAN100
!
vlan 200,300
!
vlan 333
name VLAN333
```



```
switchport mode trunk
media-type rj45
negotiation auto
!
interface GigabitEthernet1/1
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet1/2
switchport trunk encapsulation dot1q
switchport trunk native vlan 333
switchport trunk allowed vlan 20,333
switchport mode trunk
media-type rj45
negotiation auto
!
interface GigabitEthernet1/3
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet2/0
media-type rj45
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
banner exec
```

- IOSv - Cisco Systems Confidential *

-

*

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

-

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

-

*

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

banner incoming

-
- IOSv - Cisco Systems Confidential *

-

*

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

-

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

-

*

- Unauthorized use or distribution of this software is expressly *
 - Prohibited. *
- **^C
- banner login

- IOSv - Cisco Systems Confidential *

- *

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

- *

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

- *

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

!

line con 0

line aux 0

line vty 0 4

!

!

end

Switch-1#

Layer2Switc-2.log

===== PuTTY log 2025.12.15 02:17:33 =====
sh run

Building configuration...

Current configuration : 5235 bytes

!

! Last configuration change at 20:34:32 UTC Sat Dec 13 2025

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname Switch-2

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

!

!

vtp domain CISCO-VIOS

vtp mode transparent

!

!

!

ip cef

no ipv6 cef

!

!

spanning-tree mode pvst

spanning-tree extend system-id

!

vlan internal allocation policy ascending

!

vlan 20

name VLAN20

```
!  
vlan 100  
name VLAN100  
!  
vlan 200,300  
!  
vlan 333  
name VLAN333  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/1  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/2  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/3  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/0  
shutdown  
media-type rj45
```

```
negotiation auto
!
interface GigabitEthernet1/1
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet1/2
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet1/3
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet2/0
media-type rj45
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
banner exec
```

-
- IOSv - Cisco Systems Confidential *

-

*

- This software is provided as is without warranty for internal *

- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

•

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

•

*

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

banner incoming

- IOSv - Cisco Systems Confidential *

•

*

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

•

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

•

*

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

banner login

- IOSv - Cisco Systems Confidential *

- *

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

- *

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

- *

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

!

line con 0

line aux 0

line vty 0 4

!

!

end

Switch-2#

Layer2Switc-3.log

===== PuTTY log 2025.12.15 02:18:55 =====

sh run

Building configuration...

Current configuration : 5222 bytes

!

! Last configuration change at 05:53:51 UTC Sun Dec 14 2025

!


```
version 15.0
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname Switch-3
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
!
vtp domain CISCO-VIOS
vtp mode transparent
!
!
!
ip cef
no ipv6 cef
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
vlan 20
name VLAN20
!
vlan 100
name VLAN100
!
vlan 200,300
```

```
!  
vlan 333  
name VLAN333  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
switchport trunk encapsulation dot1q  
switchport trunk native vlan 333  
switchport trunk allowed vlan 20,333  
switchport mode trunk  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/1  
shutdown  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/2  
shutdown  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/3  
shutdown  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/0
```

```
switchport access vlan 20
media-type rj45
negotiation auto
!
interface GigabitEthernet1/1
switchport access vlan 333
media-type rj45
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
banner exec
```

-
- IOSv - Cisco Systems Confidential *

- *

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

- *

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

- *
- Unauthorized use or distribution of this software is expressly *
- Prohibited. *
- **^C
- banner incoming

- IOSv - Cisco Systems Confidential *

- *
- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *
- *
- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *
- *
- Unauthorized use or distribution of this software is expressly *
- Prohibited. *
- **^C
- banner login

- IOSv - Cisco Systems Confidential *

- *
- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *

- environment. *

-

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

-

*

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

!

line con 0

line aux 0

line vty 0 4

login

!

!

end

Switch-3#

Layer2Switc-4.log

===== PuTTY log 2025.12.15 02:20:09 =====

sh run

Building configuration...

Current configuration : 5222 bytes

!

! Last configuration change at 05:36:23 UTC Sun Dec 14 2025

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname Switch-4

!

```
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
!
vtp domain CISCO-VIOS
vtp mode transparent
!
!
!
ip cef
no ipv6 cef
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
vlan 20
name VLAN20
!
vlan 100
name VLAN100
!
vlan 200,300
!
vlan 333
name VLAN333
!
!
!
!
!
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
switchport trunk encapsulation dot1q  
switchport trunk native vlan 333  
switchport trunk allowed vlan 20,333  
switchport mode trunk  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/1  
shutdown  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/2  
shutdown  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet0/3  
shutdown  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/0  
switchport access vlan 20  
media-type rj45  
negotiation auto  
!  
interface GigabitEthernet1/1  
switchport access vlan 333  
media-type rj45  
negotiation auto
```

```
!  
ip forward-protocol nd  
!  
no ip http server  
no ip http secure-server  
!  
!  
!  
!  
!  
!  
control-plane  
!  
banner exec
```

-
- IOSv - Cisco Systems Confidential *

-

*

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

-

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

-

*

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

banner incoming

- IOSv - Cisco Systems Confidential *

-

*

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

-

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

-

*

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

banner login

-
- IOSv - Cisco Systems Confidential *

-

*

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

-

*

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

-

*

- Unauthorized use or distribution of this software is expressly *

- Prohibited. *

**^C

!

line con 0

line aux 0

line vty 0 4

login

!

!

end

Switch-4#

Layer2Switc-5.log

===== PuTTY log 2025.12.15 02:20:56 =====

sh run

Building configuration...

Current configuration : 5215 bytes

!

! Last configuration change at 05:36:29 UTC Sun Dec 14 2025

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname Switch-5

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

[illegible]

```
switchport trunk encapsulation dot1q
switchport trunk native vlan 333
switchport trunk allowed vlan 20,333
switchport mode trunk
media-type rj45
negotiation auto
!
interface GigabitEthernet0/1
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet0/2
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet0/3
shutdown
media-type rj45
negotiation auto
!
interface GigabitEthernet1/0
switchport access vlan 20
media-type rj45
negotiation auto
!
interface GigabitEthernet1/1
switchport access vlan 333
media-type rj45
negotiation auto
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
```

!
!
control-plane
!
banner exec

- IOSv - Cisco Systems Confidential *

- *

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

- *

- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

- *

- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

banner incoming

- IOSv - Cisco Systems Confidential *

- *

- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

- *
- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

- *
- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

banner login

- IOSv - Cisco Systems Confidential *

- *
- This software is provided as is without warranty for internal *
- development and testing purposes only under the terms of the Cisco *
- Early Field Trial agreement. Under no circumstances may this software *
- be used for production purposes or deployed in a production *
- environment. *

- *
- By using the software, you agree to abide by the terms and conditions *
- of the Cisco Early Field Trial Agreement as well as the terms and *
- conditions of the Cisco End User License Agreement at *
- <http://www.cisco.com/go/eula> *

- *
- Unauthorized use or distribution of this software is expressly *
- Prohibited. *

**^C

!

line con 0

line aux 0

line vty 0 4

!

!
end

Switch-5#